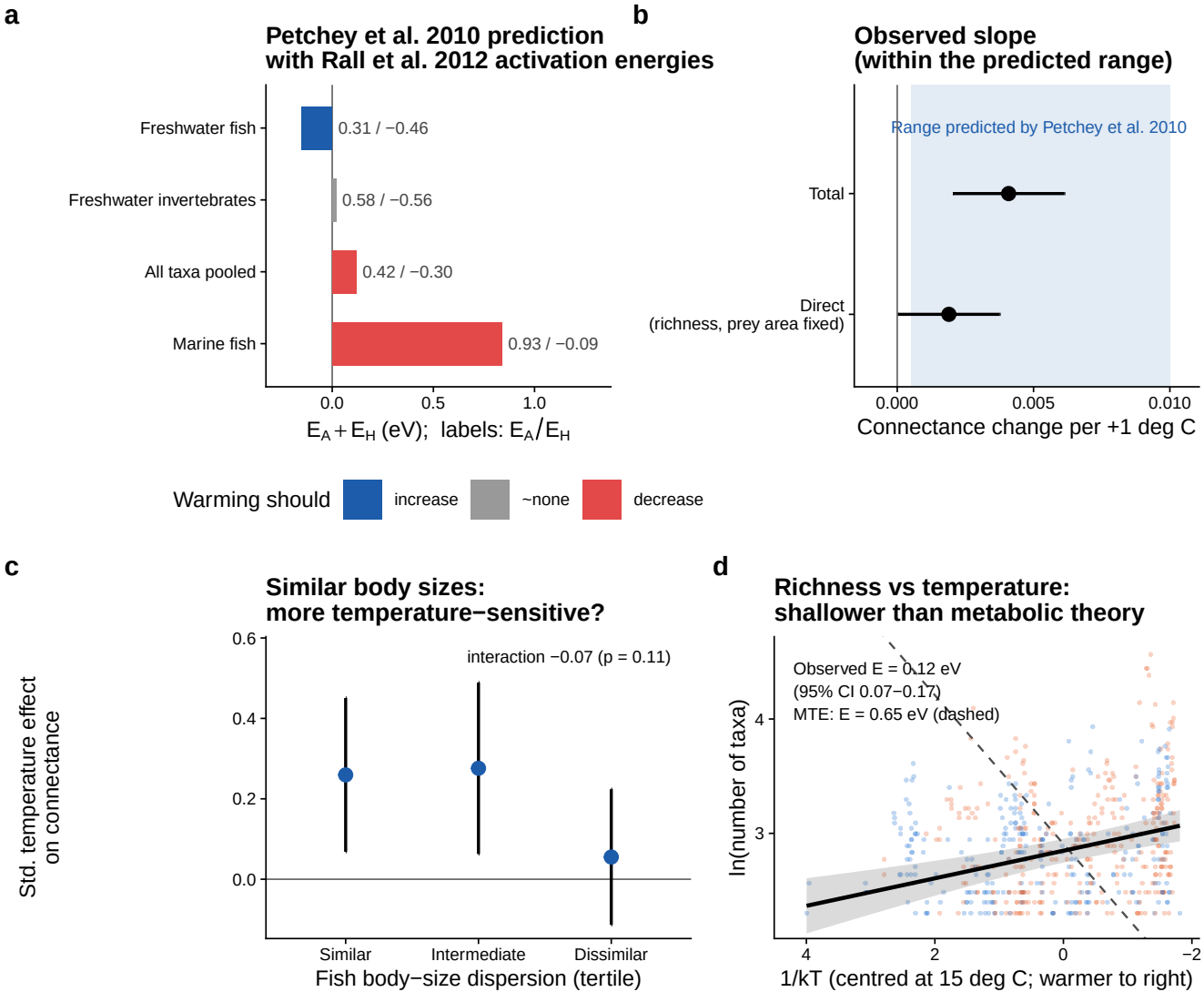
a
Petchey et al. 2010 prediction
with Rall et al. 2012 activation energies
Freshwater fish
Freshwater invertebrates
All taxa pooled
Marine fish
0.31 / −0.46
0.58 / −0.56
0.42 / −0.30
0.93 / −0.09
0.0
0.5
1.0
E_A + E_H (eV); labels: E_A/E_H
Warming should
increase
~none
decrease
b
Observed slope
(within the predicted range)
Range predicted by Petchey et al. 2010
Total
Direct
(richness, prey area fixed)
0.000
0.005
0.010
Connectance change per +1 deg C
c
Similar body sizes:
more temperature−sensitive?
interaction −0.07 (p = 0.11)
Std. temperature effect
on connectance
0.6
0.4
0.2
0.0
Similar
Intermediate
Dissimilar
Fish body−size dispersion (tertile)
d
Richness vs temperature:
shallower than metabolic theory
Observed E = 0.12 eV
(95% CI 0.07−0.17)
MTE: E = 0.65 eV (dashed)
ln(number of taxa)
4
3
4
2
0
−2
1/kT (centred at 15 deg C; warmer to right)